$$= \langle \alpha x_1 + \beta y_1, \alpha x_2 + \beta y_2, \ldots, \alpha x_n + \beta y_n, z_1, z_2, \ldots, z_n \rangle$$

$$= (\alpha x_1 + \beta y_1) z_1 + (\alpha x_2 + \beta y_2) z_2 + \ldots + (\alpha x_n + \beta y_n) z_n$$

$$= \alpha (x_1 z_1 + x_2 z_2 + \ldots + x_n z_n) + \beta (y_1 z_1 + y_2 z_2 + \ldots + y_n z_n)$$

$$= \alpha \langle u_1, v \rangle + \beta \langle u_2, v \rangle$$

I2 $\langle u, v \rangle = \sum_{i=1}^{n} x_i y_i$

$$= \sum_{i=1}^{n} y_i x_i = \langle v, u \rangle$$

I3 $\langle u, u \rangle = \sum_{i=1}^{n} x_i^2 \geq 0$

$\langle u, u \rangle = 0$

$\Leftrightarrow \sum_{i=1}^{n} x_i^2 = 0$

$\Leftrightarrow x_i = 0, \quad i = 1, 2, \ldots, n$

$\Leftrightarrow u = (0, 0, \ldots, 0)$

Since $\langle u, v \rangle : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$ st $\langle u, v \rangle = u \cdot v$ satisfy all the properties, so $\langle u, v \rangle$ is an innerproduct on $\mathbb{R}^n$.

Ex:- $\langle u, v \rangle = u^T v$, $u, v \rightarrow$ Row vector is innerproduct.

$V = \mathbb{R}^n$

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